Latency and skew

Skew is the CTS scoreboard: how different are arrival times at sinks

The idea

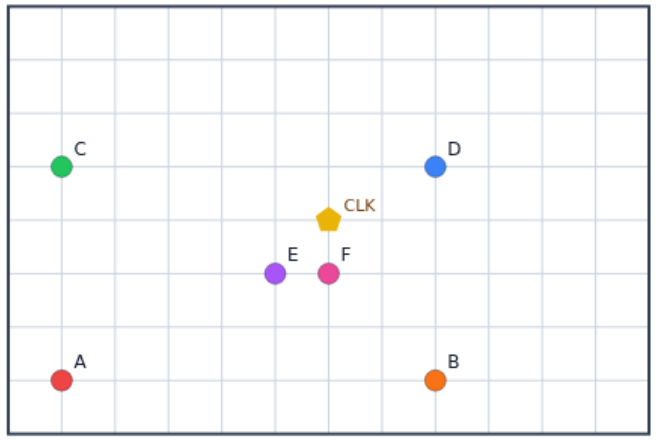
- Walk the tree from source to each sink summing Manhattan edge lengths
- Skew equals max latency minus min latency
- Wire delay per unit is one on the toy

Skew = max − min

LATENCY & SKEW

Skew = max − min

Step 1 / 5 · def · module01-03-latency-skew



Explanation

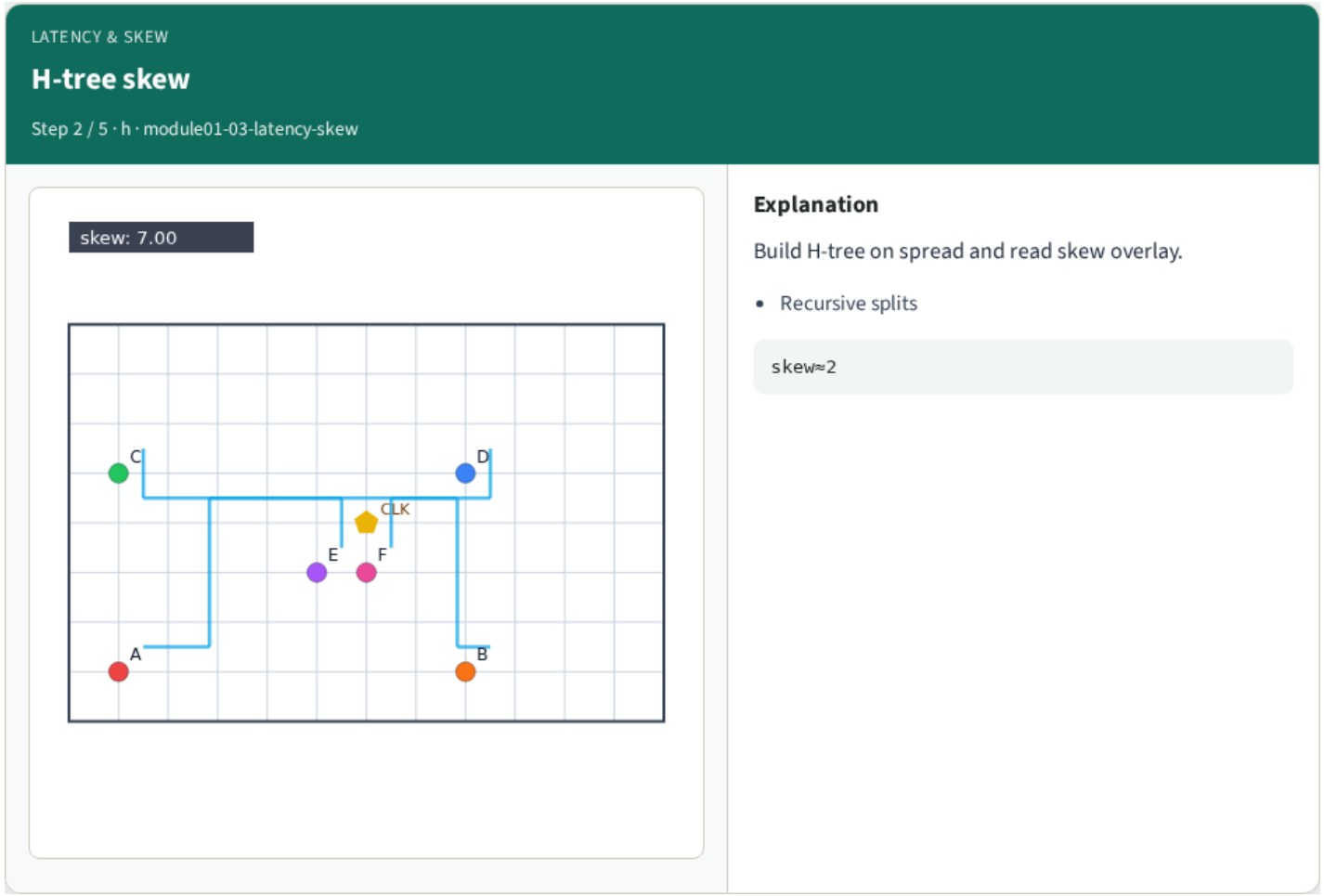
Per-sink latency is Manhattan length along the tree; skew is max minus min.

- Wire delay = 1
- Primary CTS metric

```
Document goldens
```

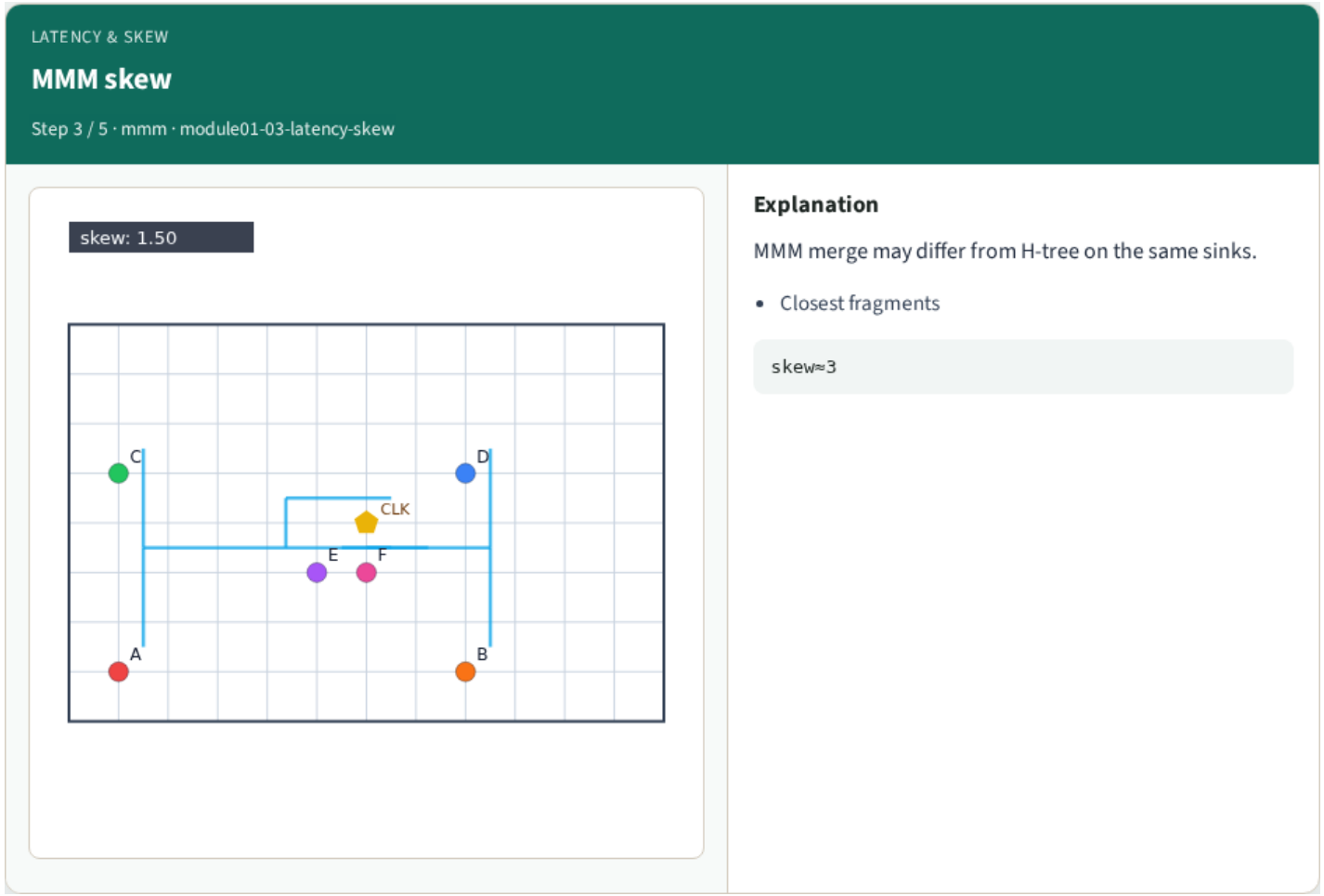
Skew = max − min

H-tree skew



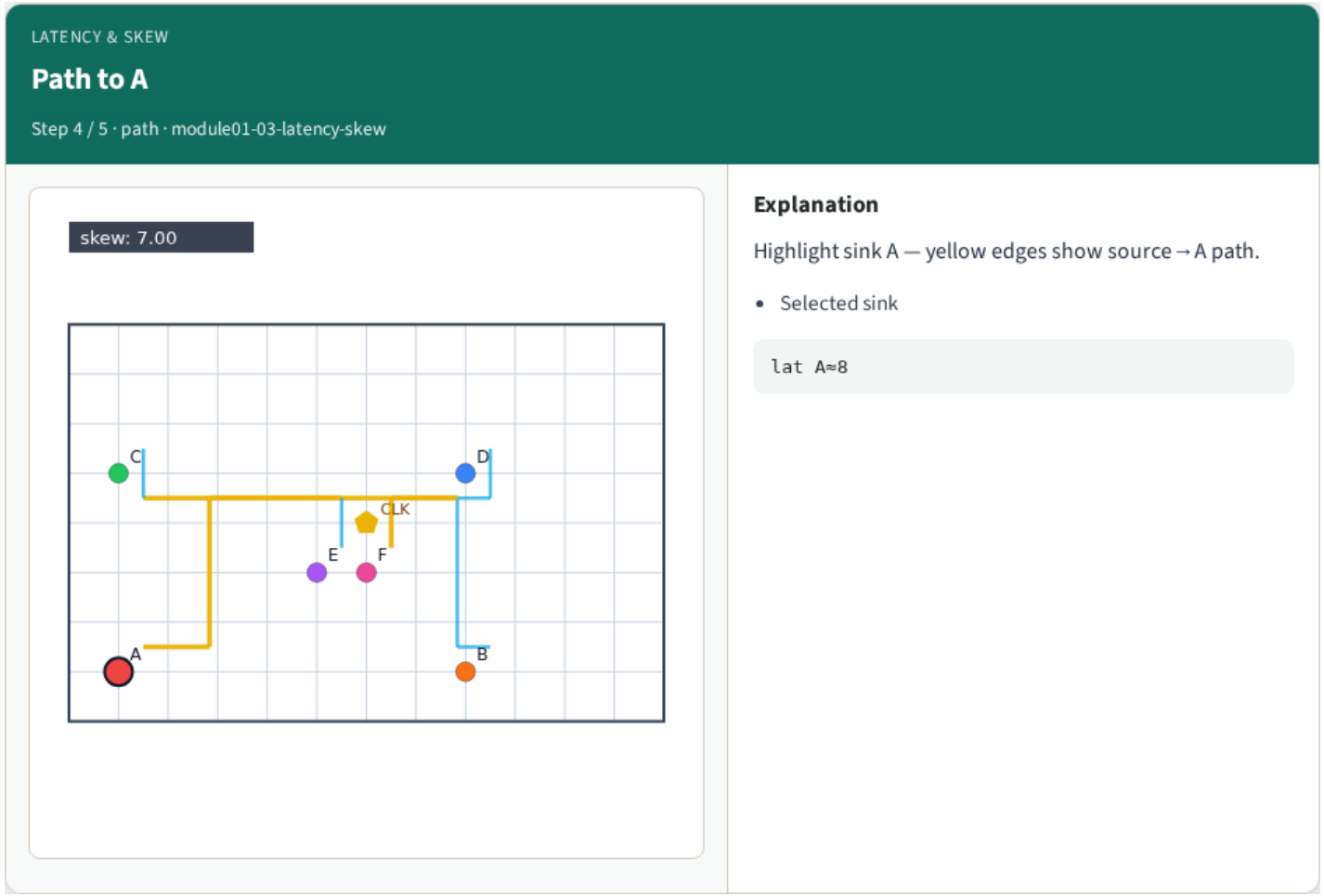
H-tree skew

MMM skew



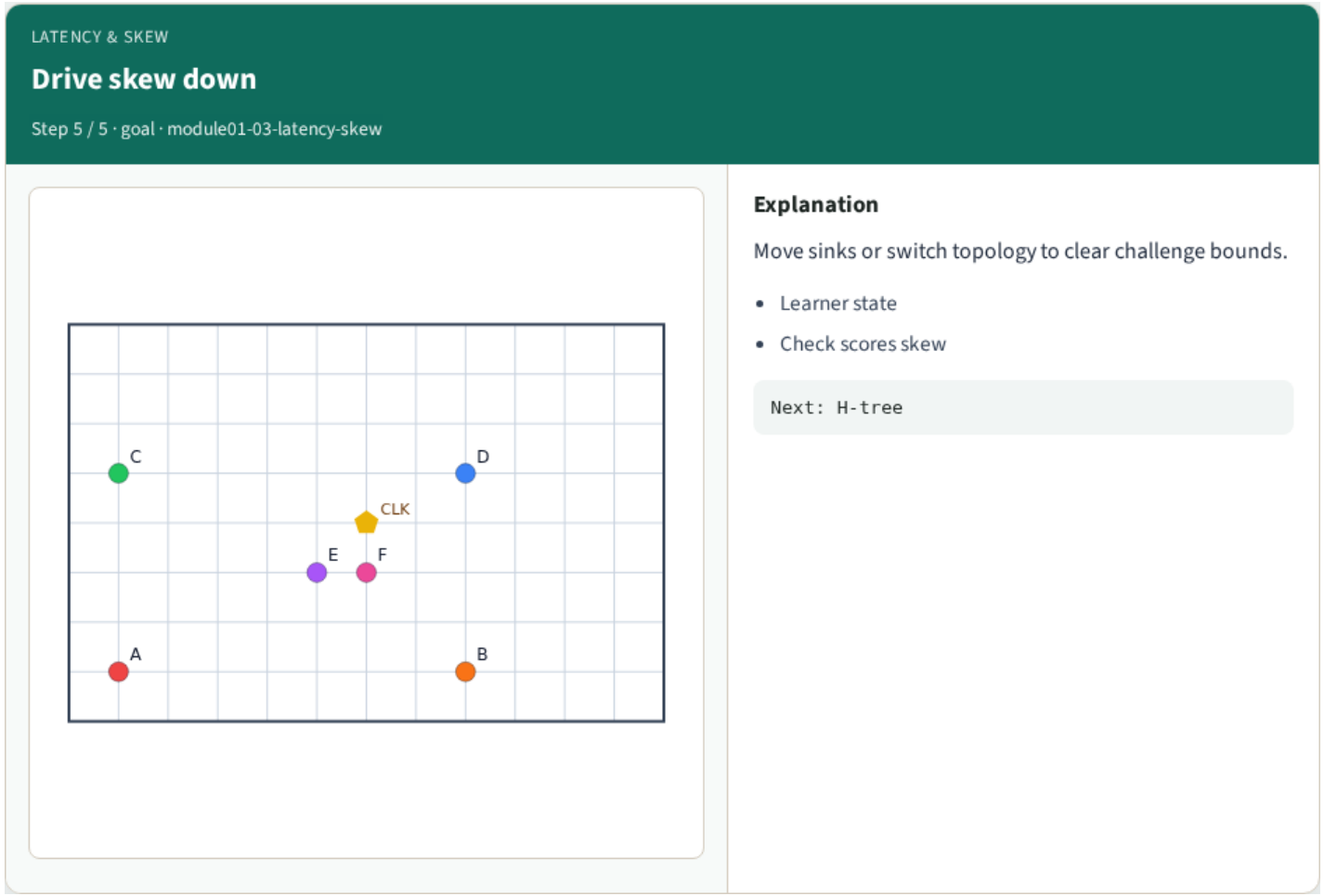
MMM skew

Path to A



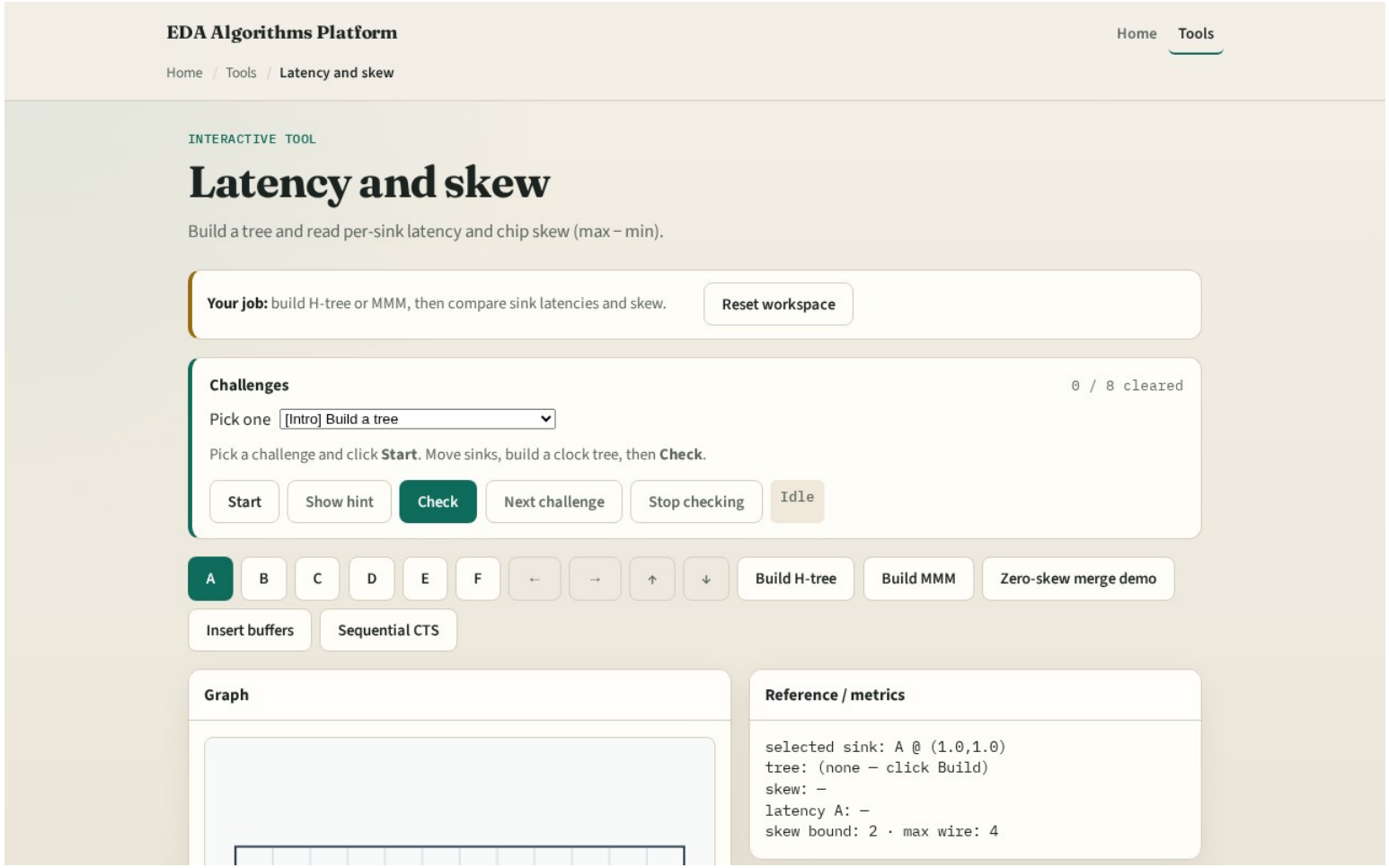
Path to A

Drive skew down



Drive skew down

Browser lab track



Browser lab starter

Implement track

- Implement or call `latency(path)` and `skew(latencies)` in `common/solvers.py`
- Print skew for MMM on tiny_cts

Pitfalls

- Using Euclidean distance
- Comparing sinks without a tree
- Treating buffer delay as wire length without counting buffers

Your turn

- Next: grow a classic H-tree topology